

MATHEMATICAL AND STATISTICAL FOUNDATIONS

B.Tech CSD

R25 Regulation | JNTU Hyderabad

UNIT-I : Basics of Number Theory

1 Greatest Common Divisors and Prime Factorization

- Greatest common divisors algorithm (Euclidean algorithm)
- The fundamental theorem of arithmetic
- Factorization of integers
- The Fermat numbers

2 Congruences

- Introduction to congruences
- Linear congmuences

Euclidean Algorithm for GCD

$$252 = 2 \times 105 + 42$$
$$105 = 2 \times 42 + 21$$
$$42 = 2 \times 21 + 0$$

$\therefore \gcd(252, 105) = 21$

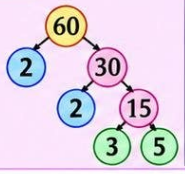
Algorithm:

1. For integers $a > b > 0$, divide $a = bq + r$ $(0 \le r < b)$.
2. Replace (a, b) by (b, r) and repeat until $r = 0$.
3. $\gcd(a, b)$ is the last non-zero remainder.

Prime Factorization

Example:

$60 = 2^2 \times 3 \times 5$



Fermat Numbers

$$F_n = 2^{2^n} + 1$$

$F_0 = 3$
$F_1 = 5$
$F_2 = 17$
$F_3 = 257$

Fundamental Theorem of Arithmetic

Every integer greater than 1 can be expressed uniquely as a product of prime numbers, apart from the order of the factors.

Example: $84 = 2^2 \times 3 \times 7$

Congruences

$$a \equiv b \pmod{m} \quad \Leftrightarrow \quad m \mid (a - b)$$

Example (Linear Congruence):

$3x \equiv 6 \pmod{9} \Rightarrow x \equiv 2 \pmod{3}$

Residues modulo 5



UNIT I : Basics of Number Theory

Q1 Definition, GCD, LCM and Prime Factorization

Question 1

Define the Greatest Common Divisor (GCD). Explain prime factorization and describe how to find the GCD and LCM of two integers using prime factors.

1. Greatest Common Divisor (GCD)

The Greatest Common Divisor (GCD) of two or more integers is the **largest positive integer** that **divides each of them exactly,** leaving remainder zero.

2. Prime Factorization

Prime factorization means expressing a positive integer as a product of **prime numbers.**

A **prime number** has exactly two positive factors : 1 and itself.

Examples : 2, 3, 5, 7, 11, 13, ...

Example : Prime factorization

For 12

$12 = 2 \times 6$

$= 2 \times 2 \times 3$

$12 = 2^2 \times 3$

For 18

$18 = 2 \times 9$

$= 2 \times 3 \times 3$

$18 = 2 \times 3^2$

3. Finding GCD using prime factorization

Example : Find GCD of 12 and 18.

$12 = 2^2 \times 3^1$, $18 = 2^1 \times 3^2$

Prime	Power in 12	Power in 18	Smaller power
2	2	1	1
3	1	2	1

$GCD = 2^1 \times 3^1$

$= 2 \times 3$

GCD(12, 18) = 6

4. Finding LCM using prime factorization

Example : Find LCM of 12 and 18.

$12 = 2^2 \times 3^1$, $18 = 2^1 \times 3^2$

Prime	Power in 12	Power in 18	Larger power
2	2	1	2
3	1	2	2

$LCM = 2^2 \times 3^2$

$= 4 \times 9$

LCM(12, 18) = 36

5. Important Formula

GCD(a, b) × LCM(a, b) = |a × b|

For 12 and 18 :

$6 \times 36 = 12 \times 18 = 216$ ✓

6. Conclusion

Using prime factorization, the GCD is obtained by multiplying the common prime factors with their **minimum exponents**, whereas the LCM is obtained by multiplying all prime factors with their **maximum exponents.**

Q2. Find the GCD and LCM of 465696 and 13608 using prime factorization.

Step 1: Prime factorization of 465696

2	465696
2	232848
2	116424
2	58212
2	29106
3	14553
3	4851
3	1617
7	539
7	77
11	11
	1

$$465696 = 2^5 \times 3^3 \times 7^2 \times 11$$

Powers of prime factors:

- 2 → 5 times
- 3 → 3 times
- 7 → 2 times
- 11 → 1 time

Step 2: Prime factorization of 13608

2	13608
2	6804
2	3402
3	1701
3	567
3	189
3	63
3	21
7	7
	1

$$13608 = 2^3 \times 3^5 \times 7$$

Powers of prime factors:

- 2 → 3 times
- 3 → 5 times
- 7 → 1 time

Step 3: Find the GCD

Prime factor	465696	13608	Take (minimum power)
2	2^5	2^3	2^3
3	3^3	3^5	3^3
7	7^2	7^1	7^1
11	11^1	–	(exclude)

$$\begin{aligned}\gcd(465696, 13608) &= 2^3 \times 3^3 \times 7 \\ &= 8 \times 27 \times 7 = 216 \times 7 \\ &= \mathbf{1512}\end{aligned}$$

Step 4: Find the LCM

Prime factor	465696	13608	Take (maximum power)
2	2^5	2^3	2^5
3	3^3	3^5	3^5
7	7^2	7^1	7^2
11	11^1	–	11^1

$$\begin{aligned}\text{LCM}(465696, 13608) &= 2^5 \times 3^5 \times 7^2 \times 11 \\ &= 32 \times 243 \times 49 \times 11 \\ &= 7776 \times 49 \times 11 \\ &= 380{,}424 \times 11 \\ &= \mathbf{4{,}184{,}664}\end{aligned}$$

Step 5: Final Answer

GCD
$= \mathbf{1{,}512}$

LCM
$= \mathbf{4{,}184{,}664}$

Verification

$\gcd(a, b) \times \text{LCM}(a, b) = a \times b$

$1512 \times 4{,}184{,}664$

$= 465696 \times 13608$

✅ Both sides are equal.

Q3 Given $mn = 2^4 \times 3^4 \times 5^3 \times 7 \times 11^3 \times 13$ and $\text{LCM}(m, n) = 2^2 \times 3^3 \times 5^2 \times 7 \times 11^2 \times 13$, find $\gcd(m, n)$.

Step 1: Formula

For any two positive integers m and n,

$$mn = \gcd(m, n) \times \text{LCM}(m, n)$$

Hence,

$$\gcd(m, n) = \frac{mn}{\text{LCM}(m, n)}$$

Step 2: Substitute the given values

$$\gcd(m, n) = \frac{2^4 \times 3^4 \times 5^3 \times 7 \times 11^3 \times 13}{2^2 \times 3^3 \times 5^2 \times 7 \times 11^2 \times 13}$$

Step 3: Simplify using laws of exponents

Use the rule $\frac{a^p}{a^q} = a^{p-q}$

Prime factor	Simplification	Result
2	2^{4-2}	2^2
3	3^{4-3}	3^1
5	5^{3-2}	5^1
7	7^{1-1}	$7^0 = 1$
11	11^{3-2}	11^1
13	13^{1-1}	$13^0 = 1$

Step 4: Final calculation

$$\begin{aligned}\gcd(m, n) &= 2^2 \times 3 \times 5 \times 11 \\ &= 4 \times 3 \times 5 \times 11 \\ &= 12 \times 5 \times 11 \\ &= 60 \times 11 \\ &= 660\end{aligned}$$

Final Answer

$$\gcd(m, n) = 660$$

Q6. Find the GCD of 2024 and 1024 using the Euclidean algorithm.

Given:
2024 and 1024

To find:
GCD(2024, 1024)

Euclidean Algorithm:
For two positive integers a and b ($a > b$),
divide a by b:
$a = bq + r$
Repeat the process with (b, r) until the remainder becomes zero.
The last nonzero remainder is the GCD.

Step-by-step solution

1	**First division:**	$2024 = 1024 \times 1 + 1000$ Remainder = 1000	gcd(2024, 1024) = gcd(1024, 1000)
2	**Second division:**	$1024 = 1000 \times 1 + 24$ Remainder = 24	gcd(1024, 1000) = gcd(1000, 24)
3	**Third division:**	$1000 = 24 \times 41 + 16$ Remainder = 16	gcd(1000, 24) = gcd(24, 16)
4	**Fourth division:**	$24 = 16 \times 1 + 8$ Remainder = 8	gcd(24, 16) = gcd(16, 8)
5	**Fifth division:**	$16 = 8 \times 2 + 0$ Remainder = 0	

Final Answer: $gcd(2024, 1024) = 8$

Q8. **Find the GCD of 252 and 198 using the extended Euclidean algorithm and express it as $252x + 198y = \gcd(252, 198)$. Find x and y.**

1. Find the GCD (Euclidean algorithm)

$$252 = 198(1) + \mathbf{54}$$
$$198 = 54(3) + \mathbf{36}$$
$$54 = 36(1) + \mathbf{18}$$
$$36 = 18(2) + \mathbf{0}$$

$$\therefore \gcd(252, 198) = \mathbf{18}$$

2. Express 18 as a linear combination of 252 and 198

Step 1: From third equation: $18 = 54 - 36$ (1)

Step 2: From second equation: $36 = 198 - 54(3)$ (2)

Substitute (2) in (1): $18 = 54 - (198 - 54(3))$
$= 4(54) - 198$ (3)

Step 3: From first equation: $54 = 252 - 198$ (4)

Substitute (4) in (3): $18 = 4(252 - 198) - 198$
$= 4(252) - 5(198)$

3. Find x and y

From $18 = 4(252) - 5(198)$

compare with $252x + 198y = 18$.

$$\mathbf{x = 4, \quad y = -5}$$

4. Verification

$$252(4) + 198(-5) = 1008 - 990$$
$$= 18$$

Q7. Find the GCD of 826 and 1890 using the extended Euclidean algorithm and express it as $826x + 1890y = \gcd(826, 1890)$. Find x and y.

Step 1: Euclidean algorithm

$$1890 = 826(2) + 238$$
$$826 = 238(3) + 112$$
$$238 = 112(2) + 14$$
$$112 = 14(8) + 0$$

The last nonzero remainder is 14.
$\therefore \gcd(826, 1890) = 14$

Final expression

$$826(-16) + 1890(7) = 14$$

Answer

$$\gcd(826, 1890) = 14$$
$$x = -16, \quad y = 7$$

Step 2: Back substitution

1. From $238 = 112(2) + 14$,
   $14 = 238 - 112(2)$

2. From $826 = 238(3) + 112$,
   $112 = 826 - 238(3)$

3. Substitute in step 1:
   $14 = 238 - 2(826 - 238(3))$
   $= 7(238) - 2(826)$

4. From $1890 = 826(2) + 238$,
   $238 = 1890 - 826(2)$

5. Substitute in step 3:
   $14 = 7(1890 - 2(826)) - 2(826)$
   $= 7(1890) - 14(826) - 2(826)$
   $= 7(1890) - 16(826)$

Q12. Calculate the first five Fermat numbers using the formula.

Fermat numbers are defined by

$$F_n = 2^{2^n} + 1$$

Find the first five Fermat numbers for $n = 0, 1, 2, 3, 4$.

1 For $n = 0$

$$\begin{aligned} F_0 &= 2^{2^0} + 1 \\ &= 2^1 + 1 \\ &= 2 + 1 \end{aligned}$$

$F_0 = 3$

2 For $n = 1$

$$\begin{aligned} F_1 &= 2^{2^1} + 1 \\ &= 2^2 + 1 \\ &= 4 + 1 \end{aligned}$$

$F_1 = 5$

3 For $n = 2$

$$\begin{aligned} F_2 &= 2^{2^2} + 1 \\ &= 2^4 + 1 \\ &= 16 + 1 \end{aligned}$$

$F_2 = 17$

4 For $n = 3$

$$\begin{aligned} F_3 &= 2^{2^3} + 1 \\ &= 2^8 + 1 \\ &= 256 + 1 \end{aligned}$$

$F_3 = 257$

5 For $n = 4$

$$\begin{aligned} F_4 &= 2^{2^4} + 1 \\ &= 2^{16} + 1 \\ &= 65536 + 1 \end{aligned}$$

$F_4 = 65537$

Final Answer

The first five Fermat numbers are

3, 5, 17, 257, 65537

n	0	1	2	3	4
2^n	1	2	4	8	16
F_n	3	5	17	257	65537

Congruence Modulo m and Its Properties

1. Definition

Two integers a and b are said to be congruent modulo m if their difference is divisible by m.

$$a \equiv b \pmod{m}$$

This means $m \mid (a - b)$.

In simple words, when a and b are divided by m, they leave the same remainder.

Example

$$17 \equiv 5 \pmod{12}$$

Because $17 - 5 = 12$, and 12 is divisible by 12.

Both numbers leave remainder 5 when divided by 12.

2. Reflexive Property

$$a \equiv a \pmod{m}$$

Every integer is congruent to itself modulo m.

Example:

$$7 \equiv 7 \pmod{5}$$

Because $7 - 7 = 0$, and 0 is divisible by 5.

3. Symmetric Property

$$a \equiv b \pmod{m} \Rightarrow b \equiv a \pmod{m}$$

If a is congruent to b modulo m, then b is congruent to a modulo m.

Example:

$$17 \equiv 5 \pmod{12}$$

Therefore, $5 \equiv 17 \pmod{12}$.

Because $5 - 17 = -12$, which is divisible by 12.

4. Transitive Property

$$a \equiv b \pmod{m} \text{ and } b \equiv c \pmod{m} \Rightarrow a \equiv c \pmod{m}$$

If a is congruent to b modulo m and b is congruent to c modulo m, then a is congruent to c modulo m.

Example:

$17 \equiv 5 \pmod{12}$ and $5 \equiv 29 \pmod{12}$

Therefore, $17 \equiv 29 \pmod{12}$.

Because $17 - 29 = -12$, which is divisible by 12.

Properties of Congruences (Modulo m)

Suppose $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$.

1 Addition Property

Rule: We can add two congruences modulo the same number.

$$a + c \equiv b + d \pmod{m}$$

Example:

$$8 \equiv 3 \pmod{5}$$
$$12 \equiv 2 \pmod{5}$$

$$8 + 12 \equiv 3 + 2 \pmod{5}$$
$$20 \equiv 5 \pmod{5}$$

Both leave remainder 0.

2 Subtraction Property

Rule: We can subtract two congruences modulo the same number.

$$a - c \equiv b - d \pmod{m}$$

Example:

$$17 \equiv 2 \pmod{5}$$
$$8 \equiv 3 \pmod{5}$$

$$17 - 8 \equiv 2 - 3 \pmod{5}$$
$$9 \equiv -1 \pmod{5}$$

Both leave remainder 4.

3 Multiplication Property

Rule: We can multiply two congruences modulo the same number.

$$ac \equiv bd \pmod{m}$$

Example:

$$8 \equiv 3 \pmod{5}$$
$$7 \equiv 2 \pmod{5}$$

$$8 \times 7 \equiv 3 \times 2 \pmod{5}$$
$$56 \equiv 6 \pmod{5}$$

Both leave remainder 1.

4 Exponentiation Property

Rule: If $a \equiv b \pmod{m}$, then for any positive integer n,

$$a^n \equiv b^n \pmod{m}$$

Example:

$$8 \equiv 3 \pmod{5}$$

$$8^2 \equiv 3^2 \pmod{5}$$
$$64 \equiv 9 \pmod{5}$$

Both leave remainder 4.

Q16. Verify $10 \equiv 4 \pmod 6$ and find the least non-negative residue of 125 modulo 9.

1 Verify $10 \equiv 4 \pmod 6$

Solution:

To verify $10 \equiv 4 \pmod 6$, we check whether 6 divides $(10 - 4)$.

$$10 - 4 = 6$$

Since 6 is divisible by 6,

$10 \equiv 4 \pmod 6$ is true.

2 Find the least non-negative residue of 125 modulo 9

Solution:

Divide 125 by 9:

$$125 = 9 \times 13 + 8$$

The remainder is 8, which is the least non-negative residue of 125 modulo 9.

Least non-negative residue = 8

Question

Find $3^{10} \bmod 7$.

Answer

We need to find the remainder when 3^{10} is divided by 7.

1 Calculate powers of 3.

$3^1 = 3$

$3^2 = 9 \equiv 2 \pmod 7$

$3^3 = 3 \times 9 = 27 \equiv 6 \pmod 7$

$3^4 = 3 \times 6 = 18 \equiv 4 \pmod 7$

$3^5 = 3 \times 4 = 12 \equiv 5 \pmod 7$

$3^6 = 3 \times 5 = 15 \equiv 1 \pmod 7$

2 Simplify 3^{10}.

Since $10 = 6 + 4$,

$$3^{10} = 3^6 \times 3^4.$$

Taking modulo 7,

$$3^{10} \equiv 1 \times 4 \pmod 7.$$

Final Answer

$$3^{10} \equiv 4 \pmod 7.$$

Q18. Linear Congruence

1. Definition

A linear congruence is an equation of the form

$$ax \equiv b \pmod{m}$$

where a, b, m are integers, $m > 0$ and x is the unknown integer.

2. Condition for a Solution

The linear congruence $ax \equiv b \pmod{m}$ has a solution if and only if

$$\gcd(a, m) \mid b$$

If $\gcd(a, m)$ divides b, then solutions exist.
The number of incongruent solutions modulo m is $\gcd(a, m)$.

3. Important Points

- If $\gcd(a, m) \mid b \Rightarrow$ solutions exist.
- Number of incongruent solutions modulo $m = \gcd(a, m)$.
- To find all solutions, solve the reduced congruence modulo m/d, where $d = \gcd(a, m)$, then add m/d repeatedly.

Example 1: Solve $4x \equiv 6 \pmod{10}$ and find all solutions

Step 1: Find the GCD

$\gcd(4, 10) = 2$

Since $2 \mid 6$, solutions exist.
Number of incongruent solutions modulo $10 = 2$.

Step 2: Divide by 2

$4x \equiv 6 \pmod{10}$

Divider by 2:

$2x \equiv 3 \pmod{5}$

Step 3: Find modular inverse

Find a number that gives remainder 1 when multiplied by 2 modulo 5.

$2 \times 3 = 6 \equiv 1 \pmod{5}$

So, inverse of 2 modulo 5 is 3.

Multiply both sides by 3:

$x \equiv 3 \times 3 \pmod{5}$

$x \equiv 9 \pmod{5}$

Since $9 = 5 + 4$,

$x \equiv 4 \pmod{5}$

Step 4: Find all solutions modulo 10

$d = \gcd(4, 10) = 2$,
so there are 2 solutions modulo 10.
Reduced modulus:

$$\frac{10}{2} = 5$$

First solution: $x_1 = 4$
Second solution: $x_2 = 4 + 5 = 9$

Step 5: Verify the solutions

For $x = 4$:
$4(4) = 16 \equiv 6 \pmod{10}$ ✓
For $x = 9$:
$4(9) = 36 \equiv 6 \pmod{10}$ ✓

Final Answer

$x \equiv 4, 9 \pmod{10}$

Example 2: No Solution

Solve $4x \equiv 3 \pmod{10}$

Step 1: Find the GCD

$\gcd(4, 10) = 2$

Step 2: Check the condition

We must have $2 \mid 3$.
But 3 is not divisible by 2.
Hence, the condition is not satisfied.

Final Answer

No solution exists
for $4x \equiv 3 \pmod{10}$.

Example 3: Single solution

Solve $7x \equiv 5 \pmod{9}$

Step 1: Find the GCD

$\gcd(7, 9) = 1$

Since $1 \mid 5$, a solution exists.
Number of solutions $= 1$.

Step 2: Find inverse of 7 modulo 9

$7 \times 4 = 28 \equiv 1 \pmod{9}$
Inverse of 7 is 4.

Step 3: Solve

$x \equiv 4 \times 5 \pmod{9}$

$x \equiv 20 \equiv 2 \pmod{9}$

Final Answer

$x \equiv 2 \pmod{9}$

Q19 Solve $3x \equiv 12 \pmod 6$ and find all incongruent solutions

1 Find the GCD

We have

$$\gcd(3, 6) = 3$$

Since $3 \mid 12$, solutions exist.

The number of incongruent solutions modulo 6 is $\gcd(3, 6) = 3$.

So, we need to find **3 solutions modulo 6**.

2 Simplify the congruence

$$3x \equiv 12 \pmod 6$$

Divide the coefficients, right-hand side and modulus by 3:

$$x \equiv 4 \pmod 2$$

Since 4 leaves remainder 0 when divided by 2,

$$x \equiv 0 \pmod 2$$

3 Find all three solutions modulo 6

The reduced modulus is

$$\frac{6}{\gcd(3, 6)} = \frac{6}{3} = 2$$

Our first solution is $x_0 = 0$.

Add 2 repeatedly to obtain three solutions:

$$x_1 = 0$$
$$x_2 = 0 + 2 = 2$$
$$x_3 = 2 + 2 = 4$$

$$x \equiv 0, 2, 4 \pmod 6$$

4 Verify the solutions

For $x = 0$

$3(0) = 0$

$0 \equiv 12 \pmod 6$ ✓

For $x = 2$

$3(2) = 6$

$6 \equiv 12 \pmod 6$ ✓

For $x = 4$

$3(4) = 12$

$12 \equiv 12 \pmod 6$ ✓

Final Answer

$$x \equiv 0, 2, 4 \pmod 6$$

Q20. Find the multiplicative inverse of 17 modulo 43 using the extended Euclidean algorithm.

Step 1: Write the condition

We need to find x such that

$$17x \equiv 1 \pmod{43}$$

Step 2: Apply the Euclidean algorithm

$$43 = 17(2) + 9$$
$$17 = 9(1) + 8$$
$$9 = 8(1) + 1$$

The last nonzero remainder is 1.

Step 3: Work backwards

$$\begin{aligned} 1 &= 9 - 8 && \text{(from } 9 = 8(1) + 1\text{)} \\ &= 9 - (17 - 9) && \text{(since } 8 = 17 - 9\text{)} \\ &= 2(9) - 17 \\ &= 2(43 - 2(17)) - 17 && \text{(since } 9 = 43 - 2(17)\text{)} \\ &= 2(43) - 4(17) - 17 \\ &= 2(43) - 5(17) \end{aligned}$$

Step 4: Find the inverse

From the last equation,

$$1 = 43(2) + 17(-5)$$

So, $17(-5) \equiv 1 \pmod{43}$

Thus, $-5 \equiv 43 - 5 \pmod{43}$

$$17^{-1} \equiv 38 \pmod{43}$$

Step 5: Verification

$$17 \times 38 = 646$$
$$646 = 43 \times 15 + 1$$

The remainder is 1,
so the answer is correct.

Final answer

The multiplicative inverse of 17 modulo 43 is

38

Q21. Solve $858x \equiv 1 \pmod 5$ using the Extended Euclidean Algorithm

Step 1: Reduce the congruence

$858x \equiv 1 \pmod 5$

$858 = 5(171) + 3$

$\Rightarrow \quad 858 \equiv 3 \pmod 5$

So, the congruence becomes

$$3x \equiv 1 \pmod 5$$

Step 2: Apply the Euclidean algorithm

$$5 = 3(1) + 2$$

$$3 = 2(1) + 1$$

The last nonzero remainder is **1.**

Step 3: Work backwards

$1 = 3 - 2$

Since $2 = 5 - 3$,

$1 = 3 - (5 - 3)$

$1 = 3 - 5 + 3$

$1 = 2(3) - 5$

Therefore,

$$1 = 2(3) + (-1)(5)$$

Step 4: Find the inverse

From $1 = 2(3) - 5$, we get

$$3(2) \equiv 1 \pmod 5$$

Therefore, the multiplicative inverse of 3 modulo 5 is **2.**

Step 5: Find x

Since $858x \equiv 1 \pmod 5$ reduces to $3x \equiv 1 \pmod 5$,

and the inverse of 3 modulo 5 is **2,**

$$x \equiv 2 \pmod 5$$

Final Answer

$$x \equiv 2 \pmod 5$$

All integer solutions are

$$x = 2 + 5k, \quad k \in \mathbb{Z}$$

Q23 Solve using Chinese Remainder Theorem

Given System of Congruences

$$x \equiv 2 \pmod 5$$
$$x \equiv 3 \pmod 7$$

Chinese Remainder Theorem

$$x \equiv \sum_{i=1}^{n} a_i M_i y_i \pmod M$$

$$M = m_1 m_2 \cdots m_n$$
$$M_i = \frac{M}{m_i}$$
$$M_i y_i \equiv 1 \pmod{m_i}$$

Step 1: Identify the values

i	a_i	m_i
1	2	5
2	3	7

Step 2: Find M

$$M = m_1 m_2 = 5 \times 7$$

$$\boxed{M = 35}$$

Step 3: Find M_1 and M_2

$$M_1 = \frac{M}{m_1} = \frac{35}{5} = 7$$

$$M_2 = \frac{M}{m_2} = \frac{35}{7} = 5$$

Step 4: Find the inverses y_1 and y_2

(ii) Find y_1 such that $7y_1 \equiv 1 \pmod 5$

$7 \equiv 2 \pmod 5 \Rightarrow 2y_1 \equiv 1 \pmod 5$

Using Euclidean algorithm:

$5 = 2(2) + 1$

$1 = 5 - 2(2) = 5 - 2(7 - 5)$

$1 = 3(5) - 2(7)$

$7(-2) \equiv 1 \pmod 5$

$$\boxed{y_1 = 3}$$

(ii) Find y_2 such that $5y_2 \equiv 1 \pmod 7$

Using Euclidean algorithm:

$7 = 5(1) + 2$

$5 = 2(2) + 1$

$1 = 5 - 2(2) = 5 - 2(7 - 5)$

$1 = 3(5) - 2(7)$

$5(3) \equiv 1 \pmod 7$

$$\boxed{y_2 = 3}$$

Step 5: Apply the CRT formula

$$x \equiv a_1 M_1 y_1 + a_2 M_2 y_2 \pmod M$$
$$x \equiv 2(7)(3) + 3(5)(3) \pmod{35}$$
$$x \equiv 42 + 45 \pmod{35}$$
$$x \equiv 87 \pmod{35}$$

Simplify:

$$87 = 35(2) + 17$$

$$\boxed{x \equiv 17 \pmod{35}}$$

Final Answer

$$\boxed{x = 17}$$

(Least positive solution)

Verification

For $x = 17$:

$17 = 5(3) + 2 \Rightarrow 17 \equiv 2 \pmod 5$

$17 = 7(2) + 3 \Rightarrow 17 \equiv 3 \pmod 7$

Both congruences are satisfied.

Question 24 : Chinese Remainder Theorem

Question Find the least positive integer that leaves remainders **2, 4, 5** and **6** when divided by **5, 6, 11** and **13,** respectively.

Step 1 : Write the congruences

$$x \equiv 2 \pmod 5$$
$$x \equiv 4 \pmod 6$$
$$x \equiv 5 \pmod{11}$$
$$x \equiv 6 \pmod{13}$$

Step 2 : Find M

$$\begin{aligned} M &= m_1 m_2 m_3 m_4 \\ &= 5 \times 6 \times 11 \times 13 \\ &= 5 \times 6 \times 143 \\ &= 30 \times 143 \end{aligned}$$

$$M = 4290$$

Step 3 : Find M_i

$$M_i = \frac{M}{m_i}$$

$$M_1 = \frac{4290}{5} = 858$$
$$M_2 = \frac{4290}{6} = 715$$
$$M_3 = \frac{4290}{11} = 390$$
$$M_4 = \frac{4290}{13} = 330$$

Step 4 : Find the inverses y_i

For $y_1 \pmod 5$

$$858y_1 \equiv 1 \pmod 5$$
$$858 \equiv 3 \pmod 5$$
$$3y_1 \equiv 1 \pmod 5$$
$$y_1 = 2$$

For $y_2 \pmod 6$

$$715y_2 \equiv 1 \pmod 6$$
$$715 \equiv 1 \pmod 6$$
$$y_2 \equiv 1 \pmod 6$$
$$y_2 = 1$$

For $y_3 \pmod{11}$

$$390y_3 \equiv 1 \pmod{11}$$
$$390 \equiv 5 \pmod{11}$$
$$5y_3 \equiv 1 \pmod{11}$$
$$y_3 = 9$$

For $y_4 \pmod{13}$

$$330y_4 \equiv 1 \pmod{13}$$
$$330 \equiv 5 \pmod{13}$$
$$5y_4 \equiv 1 \pmod{13}$$
$$y_4 = 8$$

Step 5 : Summary of values

i	a_i (remainder)	m_i (divisor)	$M_i = M/m_i$	y_i (inverse)
1	2	5	858	2
2	4	6	715	1
3	5	11	390	9
4	6	13	330	8

Step 6 : Apply CRT formula

$$x \equiv \sum_{i=1}^{4} a_i M_i y_i \pmod M$$

$$x \equiv 2(858)(2) + 4(715)(1) + 5(390)(9) + 6(330)(8) \pmod{4290}$$
$$x \equiv 3432 + 2860 + 17550 + 15840 \pmod{4290}$$
$$x \equiv 39682 \pmod{4290}$$

Step 7 : Find the least positive integer

$39682 \div 4290$

$4290 \times 9 = 38610$

$39682 - 38610 = 1072$

$$x \equiv 1072 \pmod{4290}$$

Least positive integer

$$x = 1072$$

Step 8 : Verify the answer

Divisor	Calculation	Remainder
5	$1072 = 5(214) + 2$	2
6	$1072 = 6(178) + 4$	4
11	$1072 = 11(97) + 5$	5
13	$1072 = 13(82) + 6$	6

Final Answer

$$x = 1072$$

The least positive integer that leaves remainders 2, 4, 5 and 6 when divided by 5, 6, 11 and 13 is 1072.

Q25 Application of Chinese Remainder Theorem



A child has some marbles. When grouped in **sevens, five** remain; when grouped in **elevens, six** remain; **and** when grouped in **thirteens, eight** remain. Find the **least possible number** of marbles.

1 Write the given congruences

Let the number of marbles be x.

$$x \equiv 5 \pmod 7$$
$$x \equiv 6 \pmod{11}$$
$$x \equiv 8 \pmod{13}$$

2 Apply CRT formula

$$x \equiv \sum_{i=1}^{3} a_i M_i y_i \pmod M$$

- M = product of moduli
- $M_i = M / m_i$
- y_i = inverse of M_i modulo m_i
- a_i = corresponding remainder

3 Find M and M_i

$$M = 7 \times 11 \times 13$$
$$M = 1001$$

$$M_1 = \frac{1001}{7} = 143$$
$$M_2 = \frac{1001}{11} = 91$$
$$M_3 = \frac{1001}{13} = 77$$

4 Find modular inverses

For y_1 : $143y_1 \equiv 1 \pmod 7$

$143 \equiv 3 \pmod 7 \Rightarrow 3y_1 \equiv 1 \pmod 7$

$y_1 = 5$ $(3 \times 5 \equiv 1 \bmod 7)$

For y_2 : $91y_2 \equiv 1 \pmod{11}$

$91 \equiv 3 \pmod{11} \Rightarrow 3y_2 \equiv 1 \pmod{11}$

$y_2 = 4$ $(3 \times 4 \equiv 1 \bmod 11)$

For y_3 : $77y_3 \equiv 1 \pmod{13}$

$77 \equiv 12 \pmod{13} \Rightarrow 12y_3 \equiv 1 \pmod{13}$

$y_3 = 12$ $(12 \times 12 \equiv 1 \bmod 13)$

5 Substitute in CRT formula

$$x \equiv 5(143)(5) + 6(91)(4) + 8(77)(12) \pmod{1001}$$

$5(143)(5) = 3575$
$6(91)(4) = 2184$
$8(77)(12) = 7392$

$$x \equiv 3575 + 2184 + 7392 \pmod{1001}$$
$$x \equiv 13151 \pmod{1001}$$

6 Find the least positive solution

Divide 13151 by 1001:

$$1001 \overline{)\,13151}\quad 13$$
$$-13013$$
$$138$$

$$x \equiv 138 \pmod{1001}$$

Least positive solution:

$$x = 138$$

7 Verification

Modulo 7	Modulo 11	Modulo 13
$138 = 7(19) + 5$	$138 = 11(12) + 6$	$138 = 13(10) + 8$
Remainder = 5 ✓	Remainder = 6 ✓	Remainder = 8 ✓

Final Answer

The least possible number of marbles is

138

Q27. Solve the following system of linear congruences:

$$5x + 11y \equiv 8 \pmod{13}$$
$$11x + 5y \equiv 9 \pmod{13}$$

Step 1: Simplify the congruences

$5x + 11y \equiv 8 \pmod{13}$

Since $11 \equiv -2 \pmod{13}$,

$5x - 2y \equiv 8 \pmod{13}$ ...(1)

$11x + 5y \equiv 9 \pmod{13}$

Since $11 \equiv -2 \pmod{13}$,

$-2x + 5y \equiv 9 \pmod{13}$ ...(2)

Step 2: Eliminate y

Multiply (1) by 5:

$25x - 10y \equiv 40 \pmod{13}$

$\Rightarrow 12x + 3y \equiv 1 \pmod{13}$ ...(3)

Multiply (2) by 2:

$-4x + 10y \equiv 18 \pmod{13}$

$\Rightarrow 9x + 10y \equiv 5 \pmod{13}$ ...(4)

Add (3) and (4):

$21x + 13y \equiv 6 \pmod{13}$

$\Rightarrow 8x \equiv 6 \pmod{13}$

Step 3: Find x

$8x \equiv 6 \pmod{13}$

Inverse of 8 modulo 13 is 5, since $8 \times 5 = 40 \equiv 1 \pmod{13}$.

Multiply both sides by 5:

$x \equiv 6 \times 5 \pmod{13}$

$\Rightarrow x \equiv 30 \pmod{13}$

$x \equiv 4 \pmod{13}$

Step 4: Find y

Substitute $x = 4$ in (1):

$5(4) - 2y \equiv 8 \pmod{13}$

$20 - 2y \equiv 8 \pmod{13}$

$\Rightarrow 7 - 2y \equiv 8 \pmod{13}$

$\Rightarrow -2y \equiv 1 \pmod{13}$

$\Rightarrow 11y \equiv 1 \pmod{13}$

Inverse of 11 modulo 13 is 6, since $11 \times 6 = 66 \equiv 1 \pmod{13}$.

Multiply both sides by 6:

$y \equiv 6 \pmod{13}$

$y \equiv 6 \pmod{13}$

Step 5: Verify

First equation:

$5(4) + 11(6) = 20 + 66 = 86$

$86 = 13(6) + 8 \Rightarrow$ remainder 8 ✓

Second equation:

$11(4) + 5(6) = 44 + 30 = 74$

$74 = 13(5) + 9 \Rightarrow$ remainder 9 ✓

Final Answer

$x \equiv 4 \pmod{13}$
$y \equiv 6 \pmod{13}$

The least non-negative solution is $(x, y) = (4, 6)$.

Q28. Solve the following system of linear congruences modulo 7.

1. Given equations

$$2x_1 + 5x_2 + 6x_3 \equiv 3 \pmod 7 \quad ①$$
$$2x_1 + x_3 \equiv 4 \pmod 7 \quad ②$$
$$x_1 + 2x_2 + 3x_3 \equiv 1 \pmod 7 \quad ③$$

2. Eliminate x_1 from (1) and (2)

(1) − (2):

$$(2x_1 + 5x_2 + 6x_3) - (2x_1 + x_3) \equiv 3 - 4$$
$$5x_2 + 5x_3 \equiv -1 \pmod 7$$
$$\mathbf{5x_2 + 5x_3 \equiv 6 \pmod 7} \quad ④$$

3. Eliminate x_1 from (3) and (2)

Multiply (3) by 2:

$$2x_1 + 4x_2 + 6x_3 \equiv 2 \pmod 7$$

Subtract (2):

$$(2x_1 + 4x_2 + 6x_3) - (2x_1 + x_3) \equiv 2 - 4$$
$$4x_2 + 5x_3 \equiv -2 \pmod 7$$
$$\mathbf{4x_2 + 5x_3 \equiv 5 \pmod 7} \quad ⑤$$

4. Find x_2

From (4) and (5):

$$5x_2 + 5x_3 \equiv 6 \pmod 7$$
$$4x_2 + 5x_3 \equiv 5 \pmod 7$$

Subtract (5) from (4):

$$(5x_2 - 4x_2) + (5x_3 - 5x_3) \equiv 6 - 5$$
$$\mathbf{x_2 \equiv 1 \pmod 7}$$
$$\Rightarrow x_2 = 1$$

5. Find x_3

Substitute $x_2 = 1$ in (4):

$$5(1) + 5x_3 \equiv 6 \pmod 7$$
$$5x_3 \equiv 1 \pmod 7$$

Inverse of 5 modulo 7 is 3
$(5 \times 3 = 15 \equiv 1)$

Multiply by 3:

$$x_3 \equiv 3 \pmod 7$$
$$\Rightarrow x_3 = 3$$

6. Find x_1

Use equation (2):

$$2x_1 + x_3 \equiv 4 \pmod 7$$

Substitute $x_3 = 3$:

$$2x_1 + 3 \equiv 4 \pmod 7$$
$$2x_1 \equiv 1 \pmod 7$$

Inverse of 2 modulo 7 is 4
$(2 \times 4 = 8 \equiv 1)$

Multiply by 4:

$$x_1 \equiv 4 \pmod 7$$
$$\Rightarrow x_1 = 4$$

7. Final Answer

$$\mathbf{x_1 = 4, \quad x_2 = 1, \quad x_3 = 3 \pmod 7}$$

8. Verification

(1) $2(4) + 5(1) + 6(3) = 31 \equiv 3 \pmod 7$ ✓

(2) $2(4) + 3 = 11 \equiv 4 \pmod 7$ ✓

(3) $4 + 2(1) + 3(3) = 15 \equiv 1 \pmod 7$ ✓